

MULTI-AGENT DATA ANALYSIS AND INSIGHT GENERATION PLATFORM FOR FINANCIAL AND SALES DATA

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Presentation Outline

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Motivation

- Despite the availability of data, many organizations lack the technical expertise required to analyze and interpret it effectively.
- Traditional data analysis tools often require statistical knowledge, programming skills, or manual report interpretation.
- Existing AI-based systems using LLMs may generate unverified or hallucinated insights, leading to unreliable conclusions.

Introduction

- This project proposes a Multi-Agent Data Analysis and Insight Generation Platform designed to automate the data analytics process.
- The system consists of multiple specialized AI agents that collaboratively perform data validation, cleaning, statistical analysis, visualization, insight generation, and report creation.
- Users can upload datasets in CSV format without requiring programming or analytical expertise.
- The agents work together to transform raw data into meaningful insights, visualizations, and comprehensive reports.
- The platform aims to reduce analysis time, improve accessibility, and support evidence-based decision-making through intelligent automation.

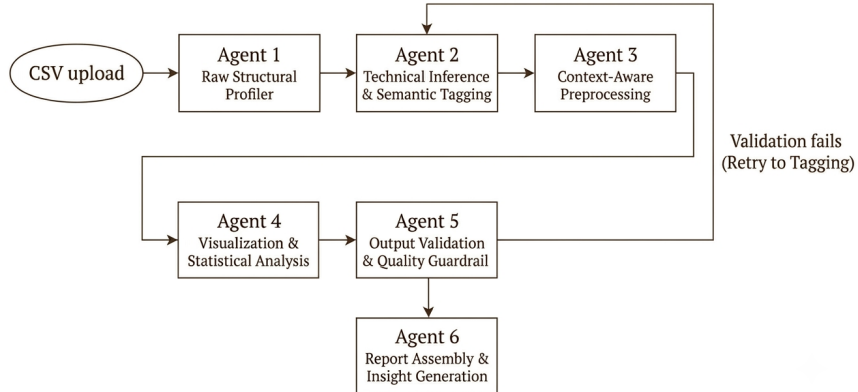
Objectives

- Develop an autonomous multi-agent system that converts raw sales and financial data (CSV format) into professional analytical reports and forecasts without manual coding.
- Democratize basic financial analytics and sales optimization for small to medium businesses and organizations through a no-code web interface.

Scope of Project

- Focus on financial and sales datasets (CSV/Excel format)
- Automated data cleaning, preprocessing, and validation
- Financial statistical analysis (profit, revenue, trends)
- Multi-agent system for structured financial analysis workflow
- Insight generation with natural language explanations
- Validation of financial insights to reduce hallucination and Report Generation

Methodology – [1] (6-Agent Orchestration Pipeline)



Agent 1 – Raw Structural Profiler

- Reads CSV file without modifying any data
- Records total number of rows and columns
- Identifies data type of each column
- Detects missing values and duplicate records
- Validates file format, date format, and currency format

Agent 2 – Technical Inference & Semantic Tagging

- Analyzes column names and sample values to understand meaning
- Classifies each column into finance categories: identifier, amount, date, category, counterparty, or sensitive
- Detects column dependencies (e.g., `is_married` only applies when `age` ≥ 18)
- Flags confidential columns such as salary, account number, and personal details
- Creates a semantic blueprint that guides all downstream agents

Agent 3 – Context-Aware Preprocessing

- Removes duplicate records detected by Agent 1
- Corrects common accounting typos and inconsistent category labels
- Converts each column to its correct data type using Agent 2's classification
- Fills missing dates using forward-fill when chronological order is preserved
- Detects and tags outliers using IQR method without removing them
- Never imputes missing financial amounts — flags them for human review instead

Agent 4 – Visualization & Statistical Analysis

- Calculates descriptive statistics
- Performs trend analysis
- Detects unusual transactions (outliers)
- Measures correlations between variables
- Generates charts and visualizations

Agent 5 – Output Validation & Quality Guardrail

- Verifies statistical calculations
- Cross-checks charts against source data
- Validates relationships and trends
- Checks whether every claim is supported by evidence
- Stops the pipeline if inconsistencies are found

Agent 6 – Report Assembly & Insight Generation

- Reads validated statistical results
- Generates human-readable explanations
- Summarizes key findings
- Creates executive summaries
- Compiles charts, insights, and statistics into PDF or HTML reports

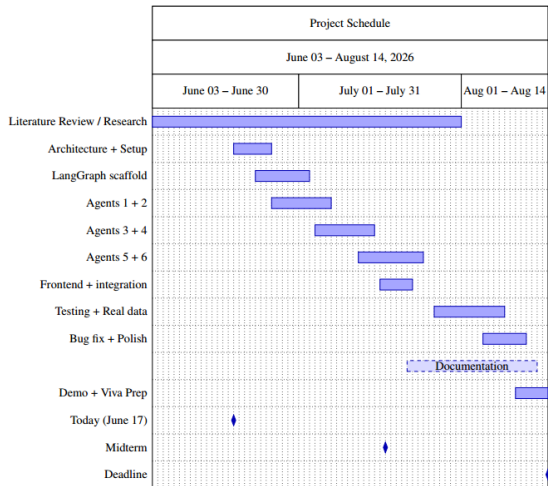
Result

- Working Dashboard
- Smart Background Processing
- Automatic PDF Reports
- No-Code Usability

Project Applications

- Financial statement analysis (profit, revenue, expenses)
- Business sales performance analysis
- ROI and profitability analysis
- Budget and expense tracking for organizations
- Banking transaction pattern analysis (basic level)
- Decision support for small and medium businesses

Gantt Chart



Analysis

- **Math vs. AI Words:** Standard math formulas handle the exact calculations while the AI handles the written explanations. This keeps the numbers 100% accurate and stops the AI from making up false statistics.
- **Fewer System Crashes:** Using LangGraph and LangSmith allows the system to catch errors and step backward to fix them instead of breaking the entire app if something goes wrong.
- **Processing Time:** Large files take a few minutes to pass through all 6 AI agents. The live status bar on the screen keeps the user updated so they are not left guessing.

Conclusion

- **Fixing Waste Data:** The system successfully proves that raw, ignored spreadsheets can be turned into valuable, easy-to-read reports automatically.
- **Beating Local Challenges:** By using free, open-source technology, it solves major real-world problems like high software costs and a lack of local tech experts in rural areas.
- **Reliable Design:** It blends rock-solid mathematical rules with creative AI explanations inside a safe, structured pipeline.

References I

1. Y. K. Dwivedi, L. Hughes, A. M. Baabdullah, et al., "So what if ChatGPT wrote it? Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy," *International Journal of Information Management*, vol. 71, 2023.
2. OpenAI, "GPT-4 Technical Report," 2024.
3. M. Sun, R. Han, B. Jiang, H. Qi, D. Sun, Y. Yuan, and J. Huang, "LAMBDA: A Large Model Based Data Agent," *arXiv preprint arXiv:2407.17535*, 2024.
4. Y. K. Liu, et al., "Summary of ChatGPT-Related Research and Perspective Towards the Future of Large Language Models," *arXiv:2306.07209*, 2023.
5. J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, "On Faithfulness and Factuality in Abstractive Summarization," in *Proceedings of ACL*, 2020.
6. V. Dibia and C. Demiralp, "Data2Vis: Automatic Generation of Data Visualizations Using Sequence-to-Sequence Recurrent Neural Networks," *IEEE Computer Graphics and Applications*, vol. 39, no. 5, pp. 33–46, 2019.
7. K. Hu, M. A. Bakker, S. Li, T. Kraska, and C. A. Hidalgo, "VizML: A Machine Learning Approach to Visualization Recommendation," in *Proceedings of the CHI Conference*, 2019.

References II

8. D. Moritz, C. Wang, G. L. Nelson, H. Lin, A. M. Smith, B. Howe, and J. Heer, “Formalizing Visualization Design Knowledge as Constraints,” *IEEE Transactions on Visualization and Computer Graphics*, 2018.
9. Y. Luo, X. Qin, N. Tang, and G. Li, “DeepEye: Towards Automatic Data Visualization,” in *IEEE ICDE*, 2018.
10. LangChain, “LangGraph: A Python Library for Building Stateful, Multi-Agent Applications,” 2024.
11. Q. Wu, G. Bansal, J. Zhang, et al., “AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation Framework,” 2023.
12. Z. Ji, et al., “Survey of Hallucination in Natural Language Generation,” *ACM Computing Surveys*, 2023.
13. G. Li, et al., “CAMEL: Communicative Agents for ‘Mind’ Exploration of Large Language Model Society,” *arXiv:2303.17760*, 2023.
14. N. Shinn, et al., “Reflexion: Language Agents with Verbal Reinforcement Learning,” *arXiv:2303.11366*, 2023.
15. D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” *ICLR*, 2015.
16. E. J. Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models,” *ICLR*, 2022.